

Digital Image Processing

Term Project #1 Report

Team 15

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Outline of Program

Purpose

The program is designed to perform basic image compositing operations, specifically addition and subtraction of images, while handling potential overflow and underflow in pixel values. This allows simple image blending and manipulation for educational purposes.

Functionality Overview

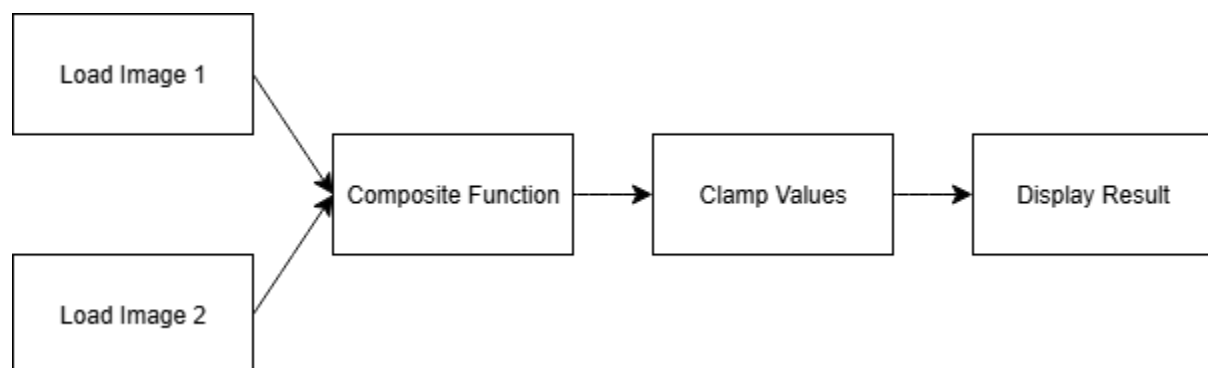
- Load input images using CxImage library
- Perform composite operations:
 - Pixel-wise addition of two images
 - Pixel-wise subtraction of one image from another
- Clamp pixel values to valid range (0–255 for 8-bit images)
- Display the resulting image

Design of Program

High-Level Design

- **Input Module:** Handles loading of images from disk
- **Composite Module:** Implements the addition and subtraction logic. Key steps include:
 1. Looping over each pixel of the input images
 2. Performing the addition or subtraction
 3. Clamping the result to avoid underflow/overflow
- **Output Module:** Displays the final image

Flow Diagram

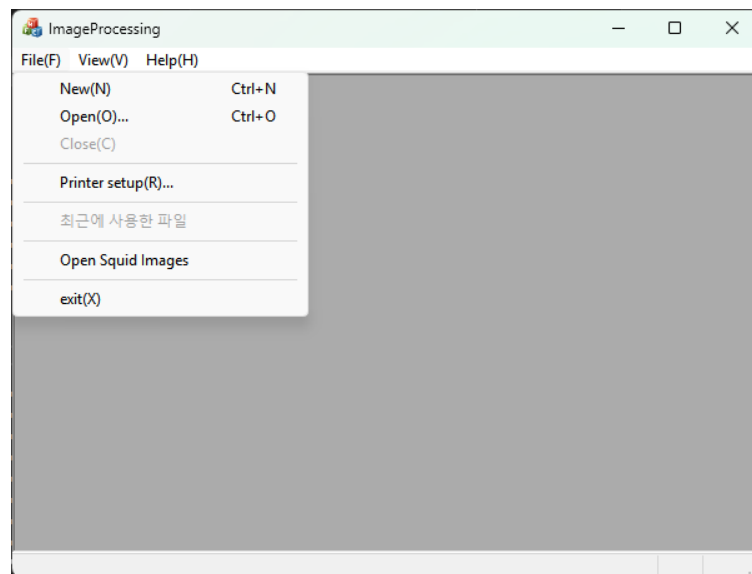


Implementation Notes

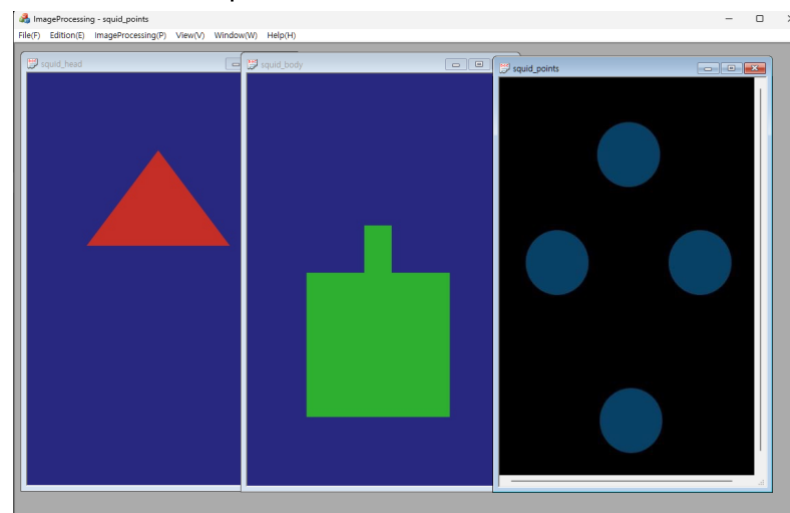
- The Composite function is implemented in a single method using simple loops over image pixels
- The program is modular, allowing easy extension for additional composite operations in the future

How to execute

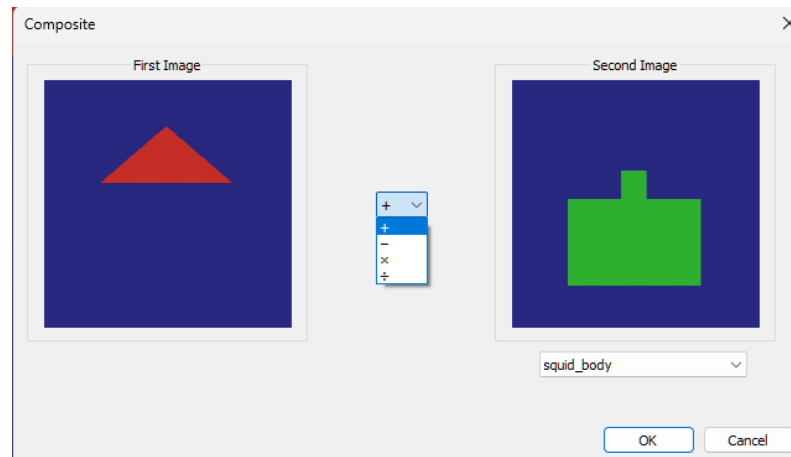
- Run the source code 'ImageProcessing.cpp'
- After the code is executed, the program runs, and a window opens
- Choose the 'File' option in the menu bar of the program as shown below



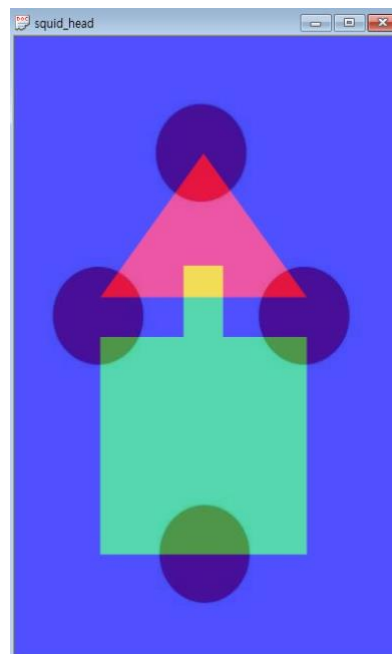
- Open all the images('squid_head.png', 'squid_body.png' and 'squid_points.png' that can found in the same directory as the ImageProcessing.cpp in folder 'Term_Project_01_Sample Images') one after another
(**Automatic way**: open all the images by choosing 'Open Squid Images option')
- All three pictures now are opened as shown below



- Click 'ImageProcessing' tab and choose 'Composite'. You can choose the second image and the operation:



- Add 'squid_body', to 'squid_head', then subtract 'squid_points' from the result of this addition
(**Automatic way:** choose 'Auto Composite Squid Images' option)
- Result is shown below



Time spent

Planning & Design: ~30 minutes

Implementation of Composite Function: ~45 minutes

Testing & Debugging: ~30 minutes

Meetings & Collaboration: ~1 hour

Total Estimated Time: ~2.5 hours

Process of Project and Record of Project Meetings

Meeting 1 – Project Kickoff

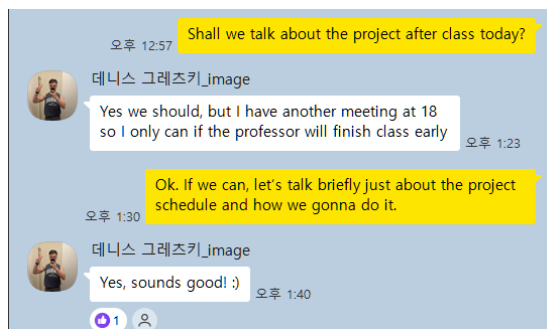
- Discussed project requirements
- Assigned tasks: one team member focused on the composite function, the other one on the clamping
- Duration: ~30 minutes

Meeting 2 – Review & Integration

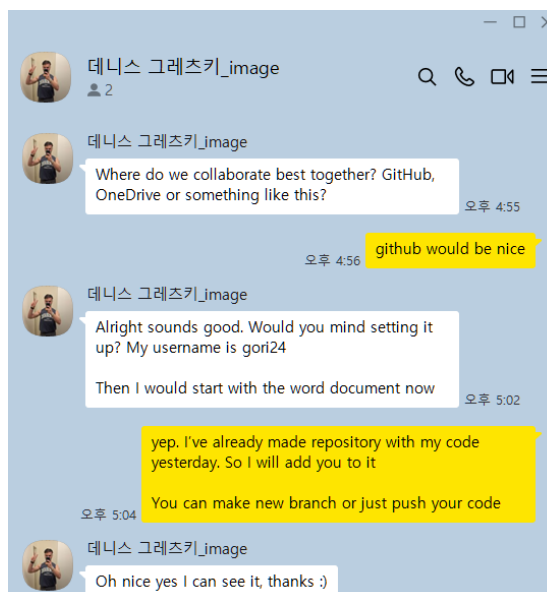
- Shared individual implementations
- Tested composite function with sample images
- Adjusted clamping logic to handle edge cases
- Duration: ~30 minutes

Records

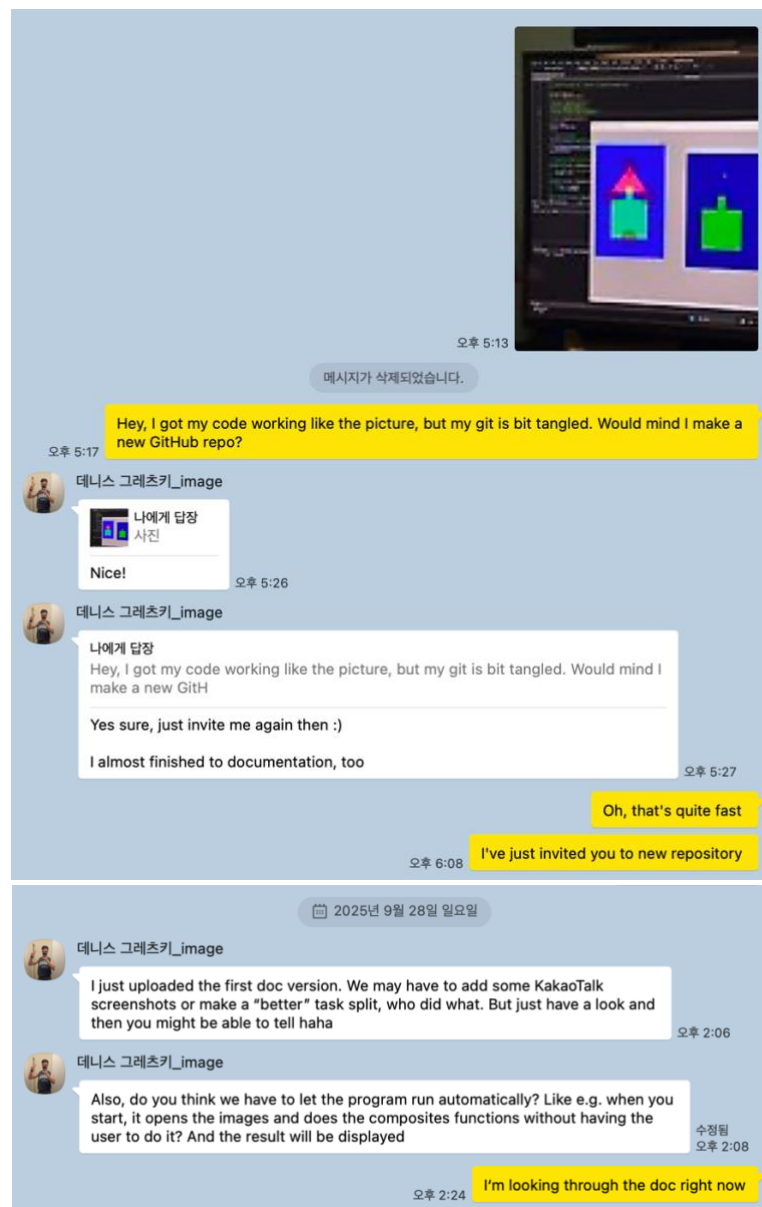
1. Adjusting schedule



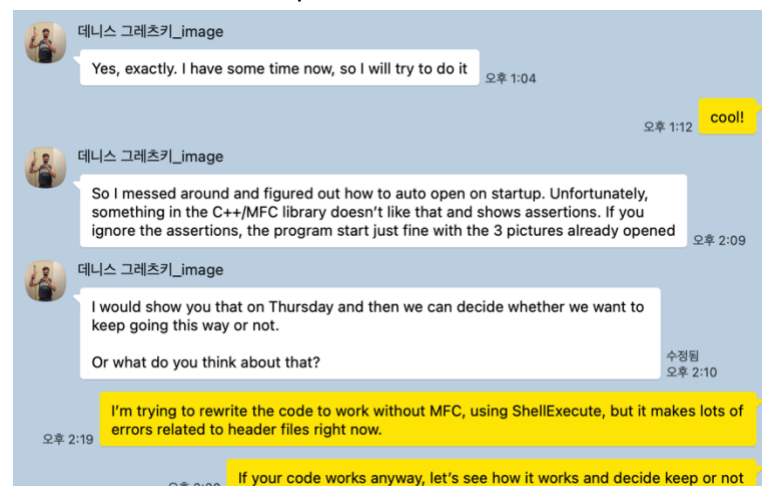
2. Discussing how to share the code



3. Cross check for executing



4. Additional process for automatic process



나에게 답장

If your code works anyway, let's see how it works and decide keep or not

on Thursday

오후 2:21



데니스 그레츠키_image

Okay yes, then lets see on Thursday :D

오후 2:31



데니스 그레츠키_image

Update: I found a way so the error does not occur. I think we can easily do the automatic way now! I will keep you updated and inform you as soon as I have a 100% reliable one

오후 3:51

오후 4:57

Great 🎉

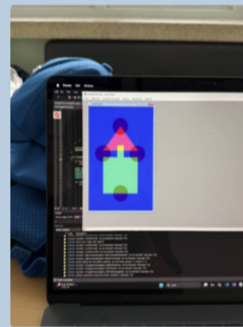
📅 2025년 9월 30일 화요일



데니스 그레츠키_image

Okay so I think it works now. Can you please try it out? Opened a new branch „automatic_processing“ for that

오후 3:36



오후 4:01

yep. I just got the result



데니스 그레츠키_image

Okay nice to see it works for you, too!

Should we include it to the doc then? And say if teacher wants to try it out manually, where he can find the files and what to do.

오후 6:15

How about ask professor tommorw about it and add how we implemented the automatic thing to the doc?

오후 7:52

📅 2025년 10월 1일 수요일



데니스 그레츠키_image

Yes we can, you want to write an email or something like that?

오전 10:47

오전 10:56

I thought it was Wednesday yesterday. I mean, ask Prof after the class



데니스 그레츠키_image

Ah okay, yes sure :D

오전 11:54



GitHub records

The screenshot displays the GitHub repository interface for 'TermProject01_DIP_Team15'. The 'Activity' tab is selected, showing a list of recent commits and branch changes. The activity is filtered by 'All activity' and 'All time'. The commits are listed in chronological order, with the most recent at the top. The commits are categorized by branch: 'master' and 'develop_Lee'. The commits include updates to project files, README.md, and the deletion of a branch. The commits are attributed to users JuhwanLee99 and Gori24. The commit messages are: 'Add project files.', 'Update README.md', 'Deleted branch', 'Add first doc version', 'Update doc V2', 'Update doc V2.0.1', 'implement automatic opening and processing', 'update doc (student number) V2.0.2', 'move pictures inside source folder', 'implement auto composite for squid images', 'change order of pictures', 'update doc (student number) V2.0.2', 'Merge branch 'automatic_processing' of https://github.com/JuhwanLee99...', and 'update docs for automatic processing'.

Activity

Showing oldest first

- Add project files.**
JuhwanLee99 created `master` · b41340e · 16 days ago
- Update README.md**
JuhwanLee99 pushed 1 commit to `master` · b41340e..6b90c9c · 16 days ago
- Add project files.**
JuhwanLee99 created `develop_Lee` · b41340e · 16 days ago
- Update README.md**
JuhwanLee99 pushed 1 commit to `develop_Lee` · b41340e..1726263 · 16 days ago
- Update README.md**
JuhwanLee99 pushed 1 commit to `develop_Lee` · 1726263..5bd94a4 · 16 days ago
- Update README.md**
JuhwanLee99 pushed 1 commit to `develop_Lee` · 5bd94a4..a43522c · 16 days ago
- Update README.md**
JuhwanLee99 pushed 1 commit to `master` · 6b90c9c..d5beb5d · 16 days ago
- Update README.md**
JuhwanLee99 created `develop_Lee` · a43522c · 16 days ago
- Deleted branch**
JuhwanLee99 deleted `develop_Lee` · 16 days ago
- Add first doc version**
Gori24 created `docs_Dennis` · 24d9cbd · 15 days ago
- Update doc V2**
JuhwanLee99 pushed 1 commit to `docs_Dennis` · 24d9cbd..44398d1 · 14 days ago
- Update doc V2.0.1**
- implement automatic opening and processing**
Gori24 created `automatic_processing` · 0f7c127 · 12 days ago
- update doc (student number) V2.0.2**
Gori24 pushed 1 commit to `docs_Dennis` · 3cebffd..5dd9b5c · 12 days ago
- move pictures inside source folder** (Force push)
Gori24 force pushed · beba189..03c3d5f · 22 hours ago
- implement auto composite for squid images**
Gori24 pushed 3 commits · 03c3d5f..c1f97a1 · 19 hours ago
- change order of pictures**
Gori24 pushed 1 commit · c1f97a1..e694191 · 17 hours ago
- update doc (student number) V2.0.2**
Gori24 pushed 1 commit · 3cebffd..5dd9b5c · 12 days ago
- Merge branch 'automatic_processing' of https://github.com/JuhwanLee99...** (Force push)
JuhwanLee99 force pushed · 5dd9b5c..da89078 · 2 days ago
- update docs for automatic processing**
Gori24 pushed 1 commit · da89078..f3d2563 · 1 hour ago

Member's Thoughts on This Project

Lessons Learned

- Even small programs benefit from modular design
- Pixel-level operations require attention to overflow/underflow
- Collaboration and code review help catch subtle mistakes

Future Improvements

- Support for blending with weights (alpha blending)
- Adding more composite modes (multiply, divide, etc.)